B 2.3.2 Selection

B 2.3.2 Construct programs utilizing appropriate selection structures.

Explore the power of if, else, and else if/elif statements. Learn how these structures, combined with Boolean and relational operators, allow programs to make decisions and execute different code blocks based on conditions. Dive into examples in Java and Python.

Learning Objectives

- Students will be able to explain the purpose and use of selection structures (if, else, else if/elif).
- Students will be able to apply Boolean and relational operators to control program flow.
- Students will be able to differentiate between if-else and multiple if statements.
- Students will be able to implement selection structures within loops and nested loops.

IB ATL Skills:

Thinking: Critical thinking (analysing and evaluating issues and ideas), Creative thinking (generating novel ideas and considering new perspectives)
Communication: Communication (exchanging thoughts and information effectively through interaction),
Self-Management: Organisation (managing time and tasks effectively),
Research: Information literacy (finding, interpreting, judging and creating information)

Lesson Resources

Presentations:

JAVA

Link to presentation

Python

Link to presentation

eBook

There are two eBooks for this section one focused on JAVA and one on Python

B 2.3.2 Java

B 2.3.2 Python

These can be used to present the content to students as an alternative to the slides. Key questions test students' understanding of the topic.

Worksheets

There are 2 worksheets for Java and Python each one has several questions on if, else, elif/else if and nested loops. You can choose to do as many as you choose in lessons. Answer keys are provided which can also be shared with students. These can be interspersed into the lesson by pausing on the slides as indicated in the Lesson Plan below.

For the Java problems, it will be worth explaining how to get input from the keyboard using a Scanner before undertaking the exercises.

B 2.3.2 Worksheet - Java

B 2.3.2 Worksheet - Python

B 2.3.2 Worksheet answers Java

B 2.3.2 Worksheet answers Python

Lesson Plan

1. Introduction (10 minutes)

Slide 1: Title Slide
Title: Selection Structures: Making Decisions in Code

Slide 2: What are Selection Structures?
Content: Explain that selection structures allow programs to make decisions and execute different code blocks based on conditions.

Slide 3: if Statement
Content: Introduce the basic if statement. Explain the syntax and its workings (condition evaluation, code execution).

2. else and else if/elif (15 minutes)

Slide 4: if-else
Content: Introduce the else block. Explain how it works with the if statement to provide alternative execution paths.

Slide 5: else if/elif
Content: Introduce else if (Java) or elif (Python). Explain how to use it for multiple conditions.

Activity 1: Predicting Code Flow (10 minutes)
Worksheet: The worksheets can be seen below:

B 2.3.2 Worksheet - Java
B 2.3.2 Worksheet - Python

Instructions: Have students work individually or in pairs.

3. Boolean and Relational Operators (15 minutes)

Slide 6: Boolean Operators
Content: Explain the use of Boolean operators (&&, ||, !) to combine conditions. Provide examples.

Slide 7: Relational Operators
Content: Review relational operators (<, >, <=, >=, ==, !=) and their role in conditions.

Activity 2: Building Conditions (10 minutes)

Worksheet: The worksheets can be seen below:

B 2.3.2 Worksheet - Java
B 2.3.2 Worksheet - Python

Instructions: Have students write the code in Java or Python, depending on the class.

4. Advanced Selection Structures (10 minutes)

Slide 8: Nested if Statements
Content: Explain how to nest if statements for more complex decision-making. Provide an example.

Slide 9: Selection within Loops

Worksheet: The worksheets can be seen below:

B 2.3.2 Worksheet - Java
B 2.3.2 Worksheet - Python

Content: Show how to use selection structures within loops to control which actions are repeated.

5. Conclusion (10 minutes)

Slide 10: Key Takeaways
Content: Summarise the different selection structures and how to use them effectively.

Slide 11: Further Exploration
Content: Provide links to online resources, tutorials, or coding challenges related to selection structures.

Assessment:

Worksheet completion: Assess student understanding based on their responses to the activity worksheets.
Class participation: Encourage active participation in discussions and problem-solving activities.